

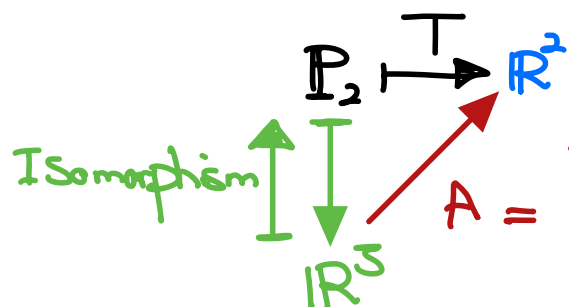
Example: Consider the transformation $\mathbb{P}_2 \xrightarrow{T} \mathbb{R}^2$ defined as

$$T(P(x)) = \begin{bmatrix} P(0) \\ P'(1) \end{bmatrix}.$$

① Using the standard basis for $\mathbb{P}_2$ and $\mathbb{R}_2$, find the matrix of T .

$$P(x) = ax^2 + bx + c \quad P'(x) = 2ax + b$$

$$T(P(x)) = \begin{bmatrix} P(0) \\ P'(1) \end{bmatrix} = \begin{bmatrix} c \\ 2a + b \end{bmatrix}$$



Isomorphism $\mathbb{P}_2 \xrightarrow{T} \mathbb{R}^2$

$\mathbb{R}^3$

$$A = \begin{bmatrix} T(\vec{e}_1) & T(\vec{e}_2) & T(\vec{e}_3) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} c \\ b \\ a \end{bmatrix} = \begin{bmatrix} c \\ 2a + b \end{bmatrix}$$

⑤ Is T one-to-one? No, since $A\vec{x} = \vec{0}$ has non-trivial solutions:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \end{array} \right] \rightarrow \vec{x} = \begin{bmatrix} c \\ b \\ a \end{bmatrix} = \begin{bmatrix} 0 \\ -2t \\ t \end{bmatrix} = t \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

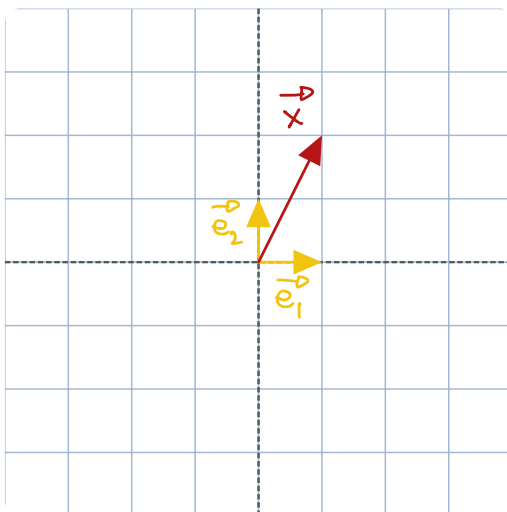
$$\left. \begin{array}{l} T(x^2 - 2x) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ T(-x^2 + 2x) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{array} \right\} \Rightarrow T \text{ is NOT one-to-one.}$$

Theorem

Let $\mathcal{B} = \{\vec{b}_i\}_{i=1}^n$ be a basis for a vector space V . Then for each $\vec{x} \in V$, there exist a unique set of scalars $\{c_i\}_{i=1}^n$ such that

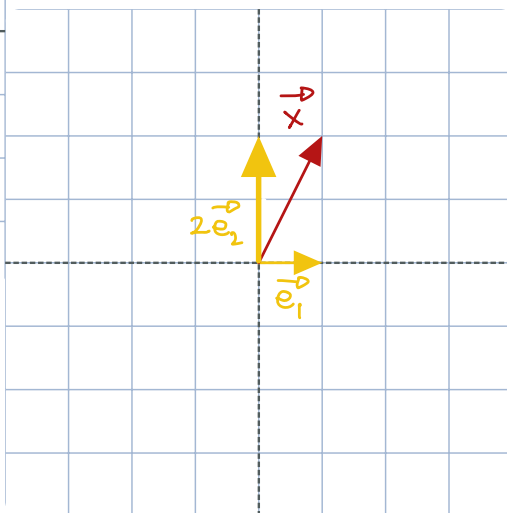
$$\vec{x} = \sum_{i=1}^n c_i \vec{b}_i.$$

The coordinates of $\vec{x}$ relative to the basis $\mathcal{B}$ is denoted by $[\vec{x}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$ and the mapping $\vec{x} \mapsto [\vec{x}]_{\mathcal{B}}$ is known as the *coordinate mapping*.



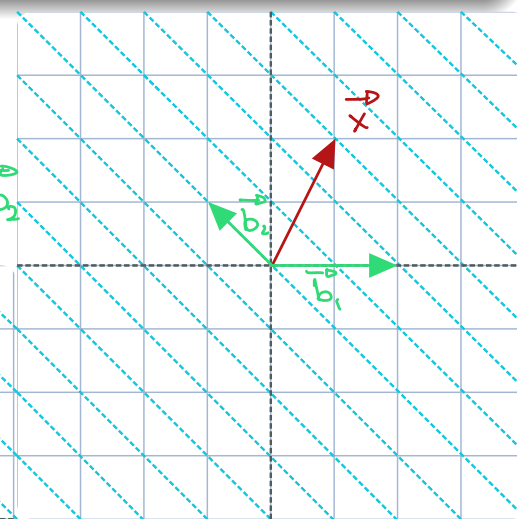
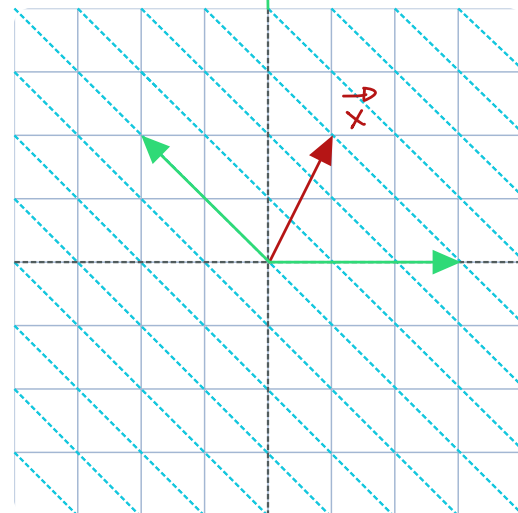
$$\mathcal{B}_1 = \{\vec{e}_1, \vec{e}_2\}$$

$$\vec{x} \mapsto [\vec{x}]_{\mathcal{B}_1} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \vec{e}_1 + 2\vec{e}_2$$



$$\mathcal{B}_2 = \{\vec{b}_1, \vec{b}_2\}$$

$$\vec{x} \mapsto [\vec{x}]_{\mathcal{B}_2} = \begin{bmatrix} \frac{1}{2} \\ 2 \end{bmatrix} = \frac{1}{2}\vec{b}_1 + 2\vec{b}_2$$



Graphical interpretation of coordinates and the change of coordinate matrix

The change of basis matrix ($P_{\mathcal{B}_2 \leftarrow \mathcal{B}_1}$):

Let $\mathcal{B}_1 = \{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n\}$ and $\mathcal{B}_2 = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ be two bases of a vector space, then to change the basis with respect to which the vector $\vec{x}$ is written from $\mathcal{B}_1$ to $\mathcal{B}_2$, we can use the change of basis matrix $P_{\mathcal{B}_2 \leftarrow \mathcal{B}_1}$ as:

$$P_{\mathcal{B}_2 \leftarrow \mathcal{B}_1} [\vec{x}]_{\mathcal{B}_1} = [\vec{x}]_{\mathcal{B}_2}$$

$$\text{where } P_{\mathcal{B}_2 \leftarrow \mathcal{B}_1} = \begin{bmatrix} [\vec{u}_1]_{\mathcal{B}_2} & [\vec{u}_2]_{\mathcal{B}_2} & \dots & [\vec{u}_n]_{\mathcal{B}_2} \end{bmatrix}.$$

In the above example, we have $\vec{e}_1 = \frac{1}{2}\vec{b}_1$ and $\vec{e}_2 = \frac{1}{2}\vec{b}_1 + \vec{b}_2$, which means $[\vec{e}_1]_{\mathcal{B}_2} = \begin{bmatrix} \frac{1}{2} \\ 0 \end{bmatrix}$, $[\vec{e}_2]_{\mathcal{B}_2} = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$ and $P_{\mathcal{B}_2 \leftarrow \mathcal{B}_1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ 0 & 1 \end{bmatrix}$

For instance:

$$\vec{x} \mapsto [\vec{x}]_{\mathcal{B}_1} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\vec{x} \mapsto [\vec{x}]_{\mathcal{B}_2} = \begin{bmatrix} \frac{3}{2} \\ 2 \end{bmatrix}$$

$$P_{\mathcal{B}_2} \leftarrow P_{\mathcal{B}_1} [\vec{x}]_{\mathcal{B}_1} = [\vec{x}]_{\mathcal{B}_2}$$

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ 2 \end{bmatrix}$$

Example: Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation given by

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -5x+6y \\ -3x+4y \end{bmatrix} \quad \begin{array}{l} \vec{e}_1 = -\vec{b}_1 + \vec{b}_2 \\ \vec{e}_2 = 2\vec{b}_1 - \vec{b}_2 \end{array}$$

and let $\beta = \{\vec{e}_1, \vec{e}_2\} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ and $\beta' = \{\vec{b}_1, \vec{b}_2\} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$ be the bases

(I) Compute matrix A_1 of T with respect to β .

$$\left. \begin{array}{l} T \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -5 \\ -3 \end{bmatrix} \\ T \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix} \end{array} \right\} \Rightarrow A_1 = \begin{bmatrix} -5 & 6 \\ -3 & 4 \end{bmatrix}$$

(II) Compute matrix A_2 of T with respect to β' .

$$\left. \begin{array}{l} T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}_{\beta'} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ T \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -4 \\ -2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} -4 \\ -2 \end{bmatrix}_{\beta'} = \begin{bmatrix} 0 \\ -2 \end{bmatrix} \end{array} \right\} \Rightarrow A_2 = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$$